

EXPLORATORY DATA ANALYSIS





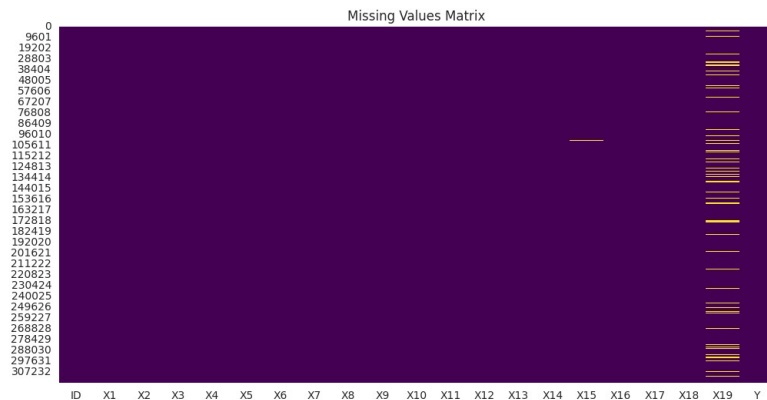
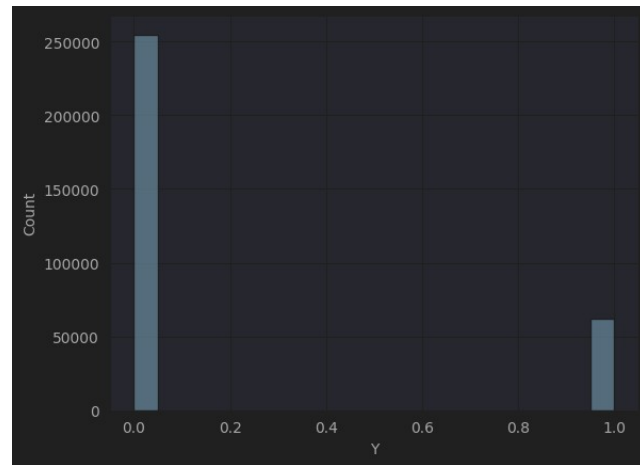
Dataset Info

~300K labelled entries

Highly imbalanced Y: 5:1 ratio

19 features, mix of categorical and numerical data.

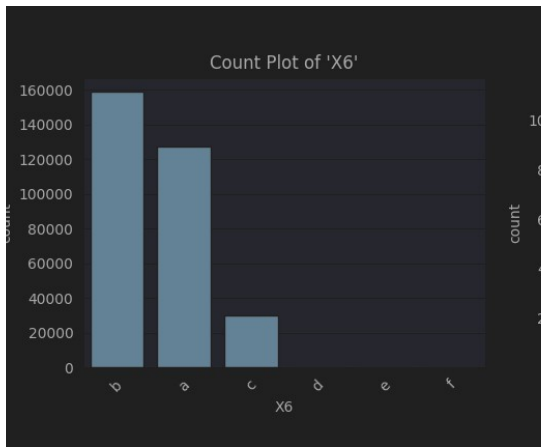
2 features missing data. ~10% X19 missing.



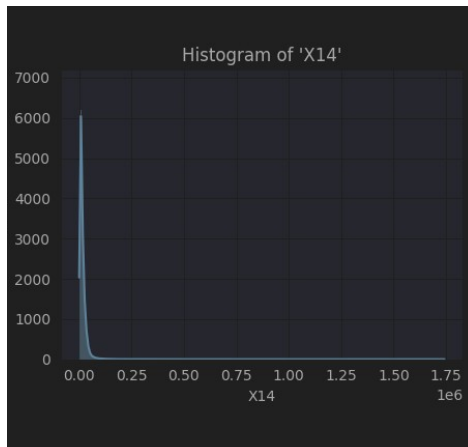


Dataset Features

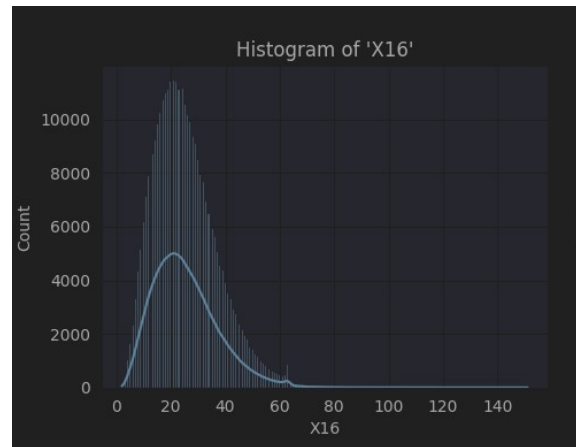
Highly imbalanced
numerical/categorical data



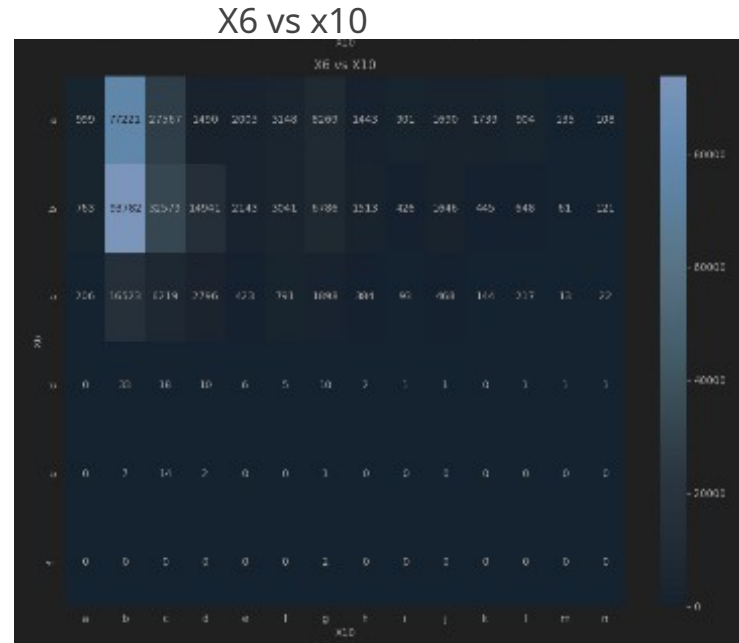
High Cluster w/ extreme outliers



Gaussian Distribution



Categorical Features - High Correlation



Same clustering of values around x10 B



Feature Engineering

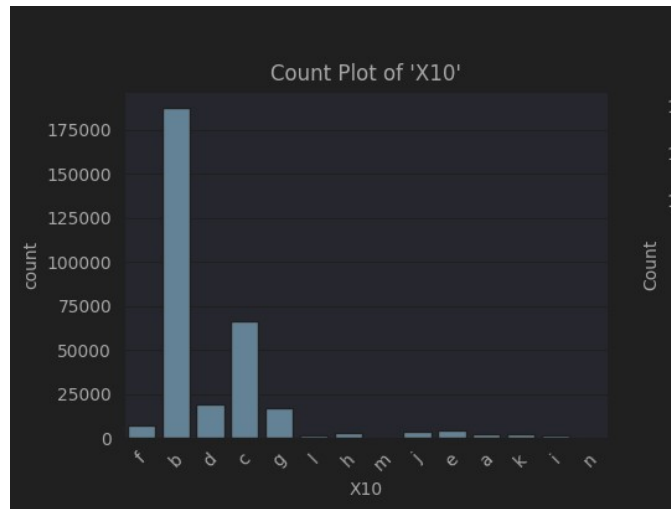
X9 dates -> No. of months since earliest date

X5, X10 -> Binarisation on most common categorical feature

X6 -> Rare features combined into a 4th "other" category

X15, X19 -> Missing data imputed with mean (due to small proportionality)

X18 -> Removed due to 99%+ of all data belonging to one category



X10 now has just 2 categories- 'b' and 'not b'



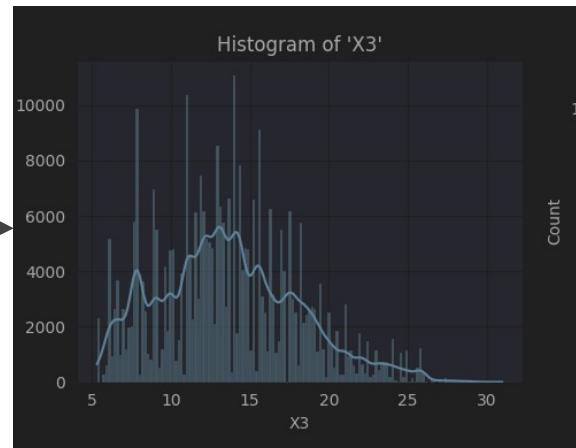
Scaling and Normalisation

X7, X11, X12, X16 (Gaussian)->
StandardScaler

X9 (Dates)-> minMax scaler

(Other numerical data)->
RobustScaler (tolerant to outliers)

Categorical data -> Binarised if 2
categories, else One Hot Encoding.



MODEL S





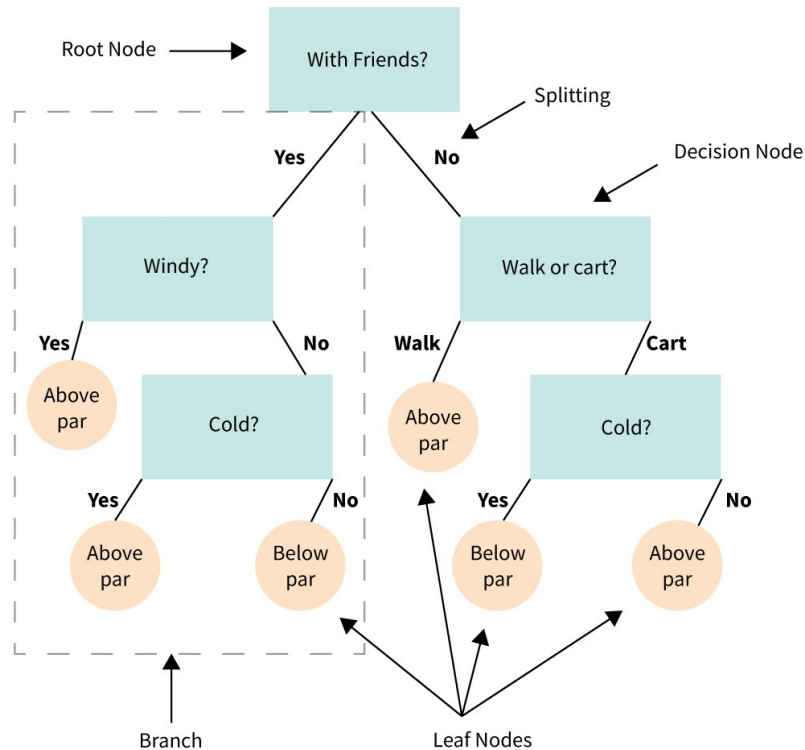
Decision Trees

Binary Yes/No Decisions.

Questions can lead to more questions, or a final conclusion.

One of the simplest classification models to exist.

Broadly applicable to many kinds of data due to its simplicity.





Decision Trees in Machine Learning

Splitting decision boundaries on data.

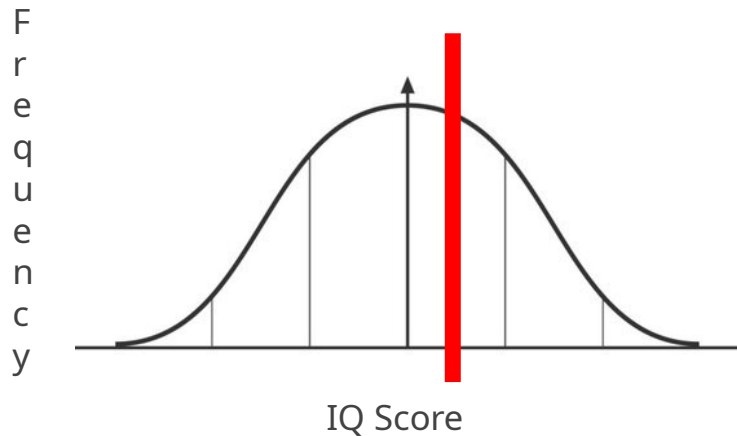
Can be used on classification and regression tasks.

Easy to apply to a variety of datasets due to its simplicity

Few assumptions about data is needed.

Often First EDA step.

Does Person X have a university degree?



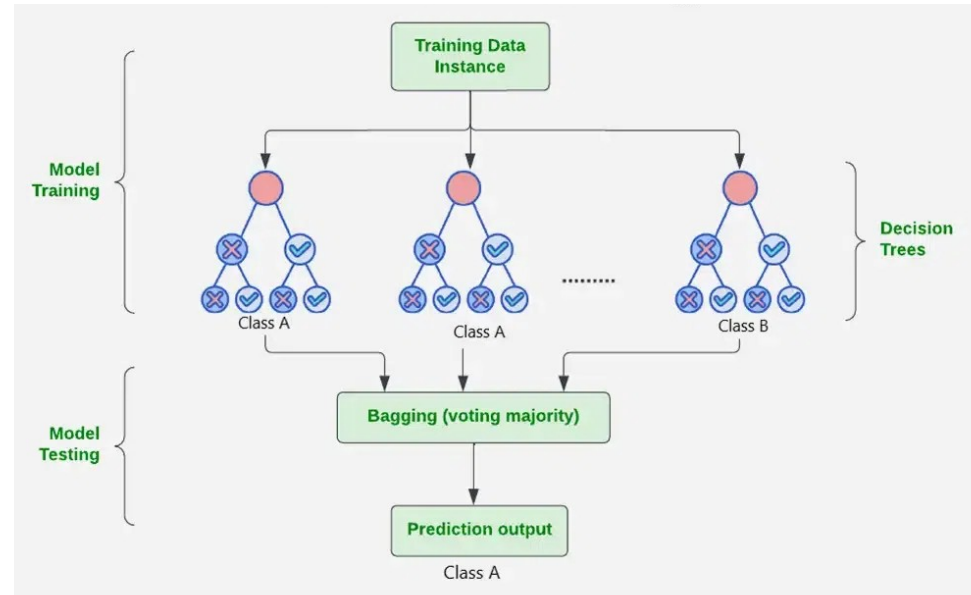
If $IQ \geq 110$, Yes. If not, No

Random Forest Algorithms

Due to stochastic randomness, trees will split on different features

Trees effectively “learn” different aspects of the same model.

Combining the results of multiple trees produces better results via majority





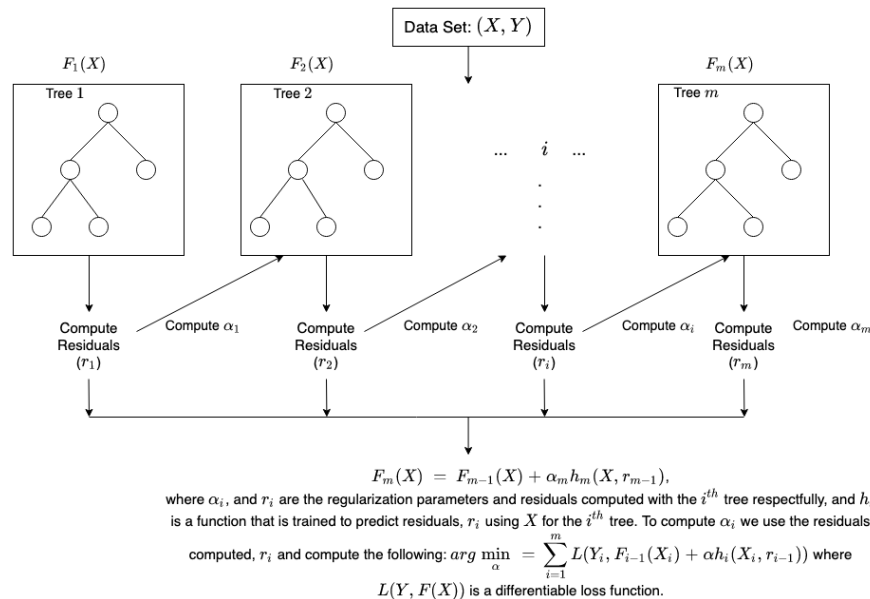
Winning model: XGBoost

XGboost is one of the best forest-based supervised learning algorithms, widely used in ML competitions.

It uses “Gradient Boosting”, which predicts a target variable using an ensemble of simpler, weaker models.

Objective function ML gradient descent.

~75% F1 score





Ensemble Modelling

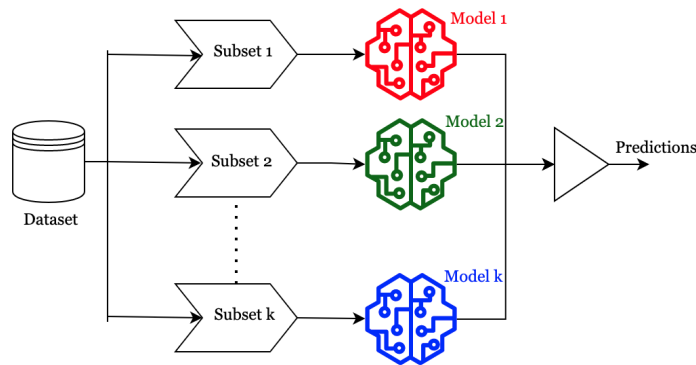
Combining multiple models together also works!

Popular method in competitions and production for reliability.

Combined 3 forest-based models:

- XGBoost (~75%)
- RandomForestClassifier (~ 74%)
- AdaBoost (~ 73%)

Worse than XGboost (74%)





Other Approaches: Neural Network

MLP Neural Network ~71%

Long time to train (~20 mins)

Used dropout layers to improve model reliability

Difficult to iterate/improve

```
nn.Linear(input_dim, 64),  
nn.ReLU(),  
nn.Linear(64, 256),  
nn.ReLU(),  
nn.Dropout(0.2),  
nn.Linear(256, 64),  
nn.ReLU(),  
nn.Dropout(0.2),  
nn.Linear(64, 8),  
nn.ReLU(),  
nn.Dropout(0.2),  
nn.Linear(8, 1),  
nn.Sigmoid()
```

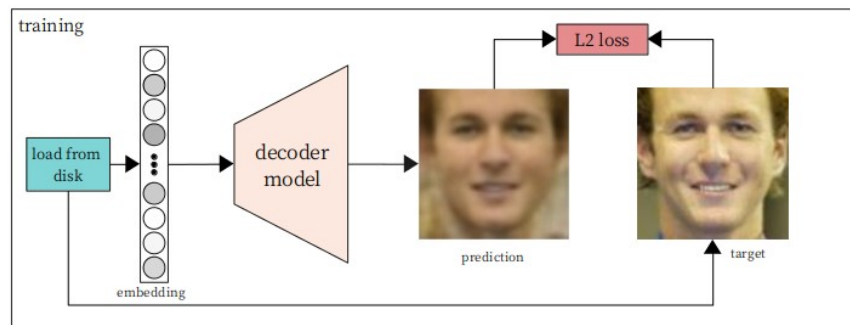
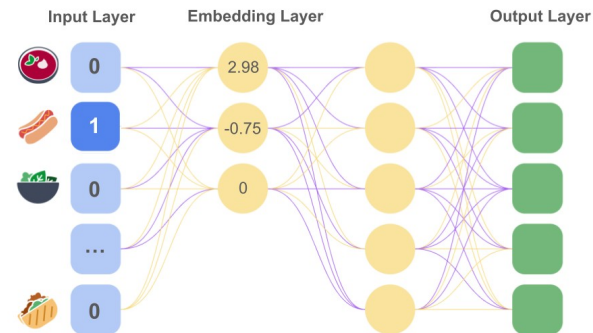
Other Approaches: Embeddings Via Encoders

Using an encoder as a dimensionality reduction technique.

Generates a small representation of a concept in larger, latent space.

Good for when there are simple features among high correlation data.

Took too long to train before I could use it (3 hours). Cannot verify if effective.



IMPROVEMENT S





Dataset Improvements

Explicitly handle outliers with caps rather than scalers

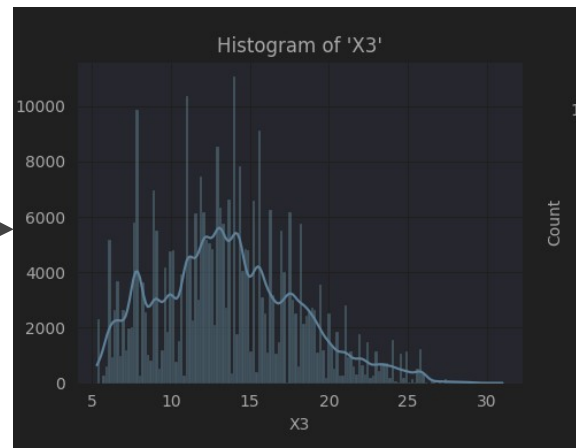
Other categorical data encoding techniques e.g Count encoding

50+ features!

Categorical data dimensionality reduction via Chi Square Testing

Using PCA or other numerical dimensionality reduction methods

Understand the origin of dataset.





Model Improvements

Encoder-based dimensionality reduction.

Non tree-based models in ensemble learning. (e.g SVM)

Other NN architectures

Exploit gaussian distribution with Gaussian Discriminant Analysis.

